

become many different kinds of specialized cells.

Sympathetic nervous system

One of the two divisions of the **autonomic nervous system**. It prepares the body for action – for example, by constricting blood vessels in the intestines and skin, widening the pupils of the eyes, and increasing the heart rate.

Synapse

The junction between two **nerve** cells, or between a **nerve** cell and a muscle fibre or a gland. Chemical messengers are passed across a synapse to produce a response in a target cell.

Synovial fluid

Thin, slippery, lubricating fluid within a joint.

Synovial joint

A mobile joint with a membrane that produces a lubricating fluid.

Systole

The period in the heartbeat cycle during which first the **atria** and then the **ventricles** contract to force blood out of the heart. See **diastole**.

T

Taste bud

A spherical nest of receptor cells found mainly on the tongue; each bud responds most strongly to a sweet, salty, sour, or bitter flavour.

Tendon

A strong band of **collagen** fibres that joins muscle to bone and transmits the pull caused by muscle contraction.

Testis (pl. testes)

One of a pair of the sperm- and **hormone**-producing sex glands in the scrotum.

Testosterone

The principal male **sex hormone**; produced in the **testis** and in small amounts in the adrenal gland on top of the kidney, and in the **ovary**.

Thalamus

A mass of **grey matter** found deep in the brain, on top of the **brainstem**. The thalamus receives and processes sensory information.

Thorax

The part of the trunk between the neck and the abdomen that contains the heart and the lungs.

Tissue

Body structure made of similar cells that perform one main function; types include muscle and connective tissue.

Tonsils

Oval masses of lymphoid tissue on the back of the throat. They help to protect against childhood infections as they attack microorganisms that enter through the nose and mouth.

Trachea

Also known as the windpipe. A muscular tube lined with **mucous membrane** and reinforced by about 20 rings of **cartilage**.

U

Umbilical cord

The structure that connects the **placenta** to the **fetus**. It provides the immunological, nutritional, and hormonal link with the mother.

Urea

A waste product of the breakdown of **proteins**; the nitrogen-containing component of urine.

Ureter

Tube through which urine passes from each **kidney** to the bladder.

Urethra

The tube that carries urine from the bladder to the exterior; much longer in the male than in the female.

Urinary tract

The system that forms and excretes urine; made up of the **kidneys**, **ureters**, bladder, and **urethra**.

Uterus

A hollow muscular structure in which the **fetus** grows until birth.

V

Vagina

The muscular passage from the **uterus** to the outside of the body; it stretches during sexual intercourse and childbirth.

Vagus nerves

The tenth pair of **cranial nerves**; helps to control automatic functions such as heartbeat and digestion.

Vas deferens

One of a pair of tubes that lead from the **testes**; each tube carries sperm, which mix with fluid before entering the **urethra**.

Vein

A thin-walled blood vessel that returns blood at low pressure from body **organs** and **tissues** to the heart.

Vena cava

One of the two large **veins**, the superior and inferior vena cavae, that empty into the right **atrium** of the heart.

Ventricle

A chamber or compartment, usually fluid-filled. Examples include two **cardiac** ventricles in the heart and four cerebral ventricles in the brain.

Vertebra (pl. vertebrae)

One of the 33 bones of the vertebral column (**spine**).

Virus

The tiniest form of infectious microorganism (germ). It takes over a cell to produce copies of itself.

Vocal cords

One of two sheets of **mucous membrane** stretched across the inside of the **larynx** that vibrate to produce voice sounds when air passes between them.

W-Z

White blood cell

Any of the colourless blood cells that play various roles in the immune system.

White matter

Nerve tissue in the brain and spinal cord formed mainly of the projecting fibres, or **axons**, of **neurons** (nerve cells).

Zygote

The single cell produced when an **ovum** is fertilized by a sperm; it contains the genetic material (**DNA**) for a new person.